

Lab-1: CS-2203 Artificial Intelligence Lab

Assignment No: 01

Date: Jan 08, 2026

Submission Deadline: [Jan 08, 2026 - 17:55 Hrs. IST]

Instructions for Submission

General Instructions

Some assignments may need to be submitted via Google Form. Please carefully follow the instructions below to ensure proper submission:

Note 1: Evaluation will start from XX.YY HRS. Please be prepared accordingly.

File Preparation

Submission Instructions:

1. Store all assignment-related files in a single folder and compress it into a `.rar` or `.zip` file. The compressed file must be named in the following format:

`roll-number_assign1.rar` or `roll-number_assign1.zip`

2. For example, if your roll number is 2201AI01, the compressed file should be named as:

`2201AI01_assign1.rar` or `2201AI01_assign1.zip`

3. Save each program using the filename format specified alongside each question.
4. If all assignments are completed within a single Jupyter Notebook, name the notebook file using the following format:

`CS2203_Lab1.ipynb`

5. When using a single notebook for multiple problems, clearly **segregate the code for each problem**. Each problem must be introduced using a **text cell** containing the problem statement before the corresponding code segment.

6. After completing the notebook, place it inside the assignment folder, compress the folder, and upload it by following the above naming instructions.
7. Additionally, submit a **PDF containing both the code and its output**. To generate the PDF:
 - Open the Jupyter Notebook file.
 - Navigate to **File** → **Print**.
 - Save the file as a PDF using the same name as the notebook (e.g., `CS2203_Lab1.pdf`).

Submission Link

Upload your compressed file using the following Google Form link:

<https://forms.gle/2t3AzjTTtW5MAatw8>

1 Introduction

Tic-Tac-Toe is a simple and popular two-player board game. In this project, an AI-based version of Tic-Tac-Toe is implemented using Python and executed in Google Colab. One player is a human, and the other player is an Artificial Intelligence (AI) agent.

The game demonstrates basic Artificial Intelligence concepts such as decision-making, rule-based logic, and opponent blocking strategies.

2 Objective of the Game

The main objective of the Tic-Tac-Toe game is:

- To place three identical symbols in a straight line.
- The line can be horizontal, vertical, or diagonal.

The human player uses the symbol **X**, while the AI uses the symbol **O**.

3 Game Board Description

- The game board
- Each cell in the grid can hold only one symbol.
- The cells are numbered from 1 to 9 for player input.

4 Tic-Tac-Toe Board Representation

Figure 1 illustrates the 3×3 Tic-Tac-Toe game board. Each cell is assigned a number from 1 to 9, which the player uses to make a move.

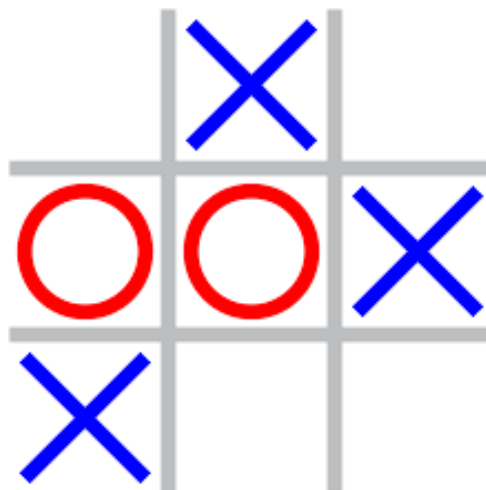


Figure 1: Tic-Tac-Toe game board

5 Explanation

- The game board consists of 9 cells arranged in a 3×3 grid.
- The human player selects a cell number (1–9) to place symbol **X**.
- The AI places symbol **O** in an available cell.
- The goal is to form a straight line of three identical symbols.

6 Rules of the Game

The following rules must be followed while playing the game:

1. The game is played between two players: a Human and an AI.
2. The human player always plays first using the symbol **X**.
3. The AI plays second using the symbol **O**.
4. A player can place their symbol only in an empty cell.
5. A player wins if they place three of their symbols in a row, column, or diagonal.
6. If all cells are filled and no player wins, the game ends in a draw.
7. Once the game is over, no further moves are allowed.

7 AI Strategy Used

The AI in this game follows a simple rule-based strategy:

- First, the AI checks if it can win in the next move.
- If winning is not possible, it checks whether the human player can win and blocks that move.
- If neither condition is met, the AI selects a random empty position.

This approach allows the AI to play intelligently without using complex machine learning algorithms.

Program: Human vs AI Tic-Tac-Toe

Language: Python

```
1 import random
2
3 # Initialize board
4 board = [" " for _ in range(9)]
5
6 def print_board():
7     print()
```

```

8     print(board[0], "|", board[1], "|", board[2])
9     print("---+---+---")
10    print(board[3], "|", board[4], "|", board[5])
11    print("---+---+---")
12    print(board[6], "|", board[7], "|", board[8])
13    print()
14
15    def check_winner(player):
16        win_conditions = [
17            (0,1,2), (3,4,5), (6,7,8),
18            (0,3,6), (1,4,7), (2,5,8),
19            (0,4,8), (2,4,6)
20        ]
21        return any(board[a] == board[b] == board[c] == player
22                    for a, b, c in win_conditions)
23
24    def is_draw():
25        return " " not in board
26
27    def human_move():
28        while True:
29            move = int(input("Enter your move (1-9): ")) - 1
30            if board[move] == " ":
31                board[move] = "X"
32                break
33            else:
34                print("Invalid move, try again.")
35
36    def ai_move():
37        # Try to win
38        for i in range(9):
39            if board[i] == " ":
40                board[i] = "O"
41                if check_winner("O"):
42                    return
43                board[i] = " "
44
45        # Try to block human
46        for i in range(9):
47            if board[i] == " ":
48                board[i] = "X"
49                if check_winner("X"):
50                    board[i] = "O"
51                    return
52                board[i] = " "
53
54        # Random move
55        move = random.choice([i for i in range(9) if board[i] == " "])
56        board[move] = "O"
57
58    # Game loop
59    print("Tic-Tac-Toe: You (X) vs AI (O)")

```

```

60 print_board()
61
62 while True:
63     human_move()
64     print_board()
65     if check_winner("X"):
66         print("You win!")
67         break
68     if is_draw():
69         print("It's a draw!")
70         break
71
72     ai_move()
73     print("AI move:")
74     print_board()
75     if check_winner("O"):
76         print("AI wins!")
77         break
78     if is_draw():
79         print("It's a draw!")
80         break

```

8 Explanation of Functions

8.1 Board Initialization

```
board = [" " for _ in range(9)]
```

This line initializes the Tic-Tac-Toe board as a list of 9 empty cells, representing a 3×3 grid.

8.2 print_board()

This function displays the current state of the game board in a readable 3×3 format. It helps the player visualize the game after each move.

8.3 check_winner(player)

This function checks whether the given player (either **X** or **O**) has won the game. It compares the board positions against all possible winning combinations, including rows, columns, and diagonals.

8.4 is_draw()

This function determines whether the game has ended in a draw. If all board positions are filled and no player has won, the function returns true.

8.5 human_move()

This function takes input from the human player. The player selects a cell number between 1 and 9. The function ensures that the selected cell is empty before placing the symbol **X**.

8.6 ai_move()

This function implements the AI logic:

- First, the AI checks if it can win in the next move.
- If not, it checks whether the human player can win and blocks that move.
- If neither condition is satisfied, the AI selects a random empty cell.

This rule-based strategy allows the AI to make intelligent decisions without using machine learning algorithms.

9 Conclusion

This AI-based Tic-Tac-Toe game is a simple yet effective example of how Artificial Intelligence can be applied to decision-making problems. It is suitable for beginners to understand AI logic, game rules, and basic programming concepts. The implementation can be extended further using advanced AI techniques such as the Minimax algorithm or Reinforcement Learning.

Output: Human vs AI Tic-Tac-Toe Game

```
... Tic-Tac-Toe: You (X) vs AI (O)

| | 
--+-+--
| | 
--+-+--
| | 

Enter your move (1-9): 1
```

(a) Initial game board and first human move

```
X | | 
--+-+--
| | 
--+-+--
| | 
```

(b) Game board after human move

```
AI move:

X | | 
--+-+--
| | O
--+-+--
| | 

Enter your move (1-9): 7
```

(c) AI move and updated board state

```
Enter your move (1-9): 9

X | | O
--+-+--
O | X | O
--+-+--
X | | X

🎉 You win!
```

(d) Final board showing human victory

Figure 2: Execution output of the Human vs AI Tic-Tac-Toe game

Evaluation Criteria

The evaluation of this lab assignment will be based on the following criteria:

1. **Correctness of Implementation (40%):**
2. **Code Quality & Q&A (30%):**
 - The code should be well-structured, properly commented, and follow standard Python programming practices.
 - Students may be asked questions related to their implementation to assess their understanding.
3. **Error Handling (10%):**
 - The code gracefully handles potential errors, such as file not found or invalid image formats.
4. **Documentation (10%):**
 - The code is accompanied by appropriate documentation and comments that explain the steps taken in each part of the program.
5. **Timeliness of Submission (10%):**
 - The assignment is submitted on time, adhering to the submission deadline.